

Ismaeel Bashir

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Skills Summary

Languages: Python, C++, Java, C, SQL, C#

Skills: Software development, Machine Learning, Data analytics, Data modelling, Systems engineering, Software engineering.

Interests/Hobbies: MMA, Poker

Education

MASTER OF INFORMATICS (MINF) | 2021-2026 | UNIVERSITY OF EDINBURGH

- Expected **First Class Honors (86%)**.
- Bsc Thesis: Training Machine Learning Models on Edge Devices **84% (published to MobiUK'25 and waiting for NSDI'25)**
- Notable Courses: Computational Logic, OOP, Linear Algebra, Calculus, Discrete Mathematics, Computer Systems, Data Science, Algorithms & Data Structures, NLP, Software Testing, Compiling Techniques, Computer Security, Computer Architecture, OS and ML Systems, Message Passing Programming, Performance Programming,

Experience

Qube Research Technologies | Quantitive Developer Intern | Python, Pandas, Numpy | Jun 2025 – Sept 2025

- Joined the HFT Quant Developer Research team. Worked on optimising research pipelines. Created a new dynamic scheduler that **improves pipeline runtime by 400%**. Automated back testing strategies and scaled research platform to handle **1-million+ tasks**. Built a cache for research pipelines, while optimising the HPC object store to **instantly re-compute ML models**.

EPCC | Researcher | C++, MPI, Python, Lustre | Jun 2025 – Present

- Working under the Director of Research to lead a research project on **profiling, optimising and scheduling ML workloads on HPC clusters**. The project is a **collaboration with QRT** that will conclude with a **first author whitepaper**.

Morgan Stanley | Software Engineer Intern | Java, Spring | Jun 2024 – Aug 2024

- Designed and created a new scalable data feed for a trading data pipeline that handles over **\$900 Bn** worth of assets.
- Created batch processes to calculate how much of a client's assets could be used for trading when on margin, **increased number of assets traded daily by \$400k**.
- Found a memory optimisation in the system that led to a **10% increase** in startup of the entire trading system.

ICSA Edinburgh University | Research Assistant | C++, GRPC, Torch | Sept 2023 – May 2025

- Worked on building low latency, low level systems using C++ for researchers studying machine learning systems. Used GRPC to setup communication channels between servers, and researched efficient tensor operations on edge devices. Work was written in my thesis and published to MobiUK'25 (waiting for NSDI'25 acceptance).

Optiver, Citadel, DE Shaw | Spring Insight Intern | May 2024

- Selected from **4000+ (1.7% of people)** for DE Shaw, Citadel, and Optiver's spring insights.
- Took part in quantitative and algorithmic trading challenges while also being informed of what role technology has in trading.
- Learned about how market makers use efficient serialization to communicate between servers.

Publications

- Second Author:** On Harnessing Idle Compute at the Edge for Foundation Model Training (**MobiUK'25**, pending NSDI'25)

Projects

Optiver Hackathon – First Place | Python | Quantitive Development |

- Built a high frequency pairs trading model for a hackathon sponsored by Optiver to make the most money trading ETF's and their stocks. Achieved **first place** by optimizing ETF and stock trading.

Marshall Wace Hackathon – First Place | Python, Java | Streamed Data Pipelining |

- Built a streamed data pipeline to allow hedge funds to send high frequency data to traders and investors.
- Received **first place** in the hackathon for having a system that was able to scale to millions of users and requests at once.

Orderbook Matching Engine | C++ | Low latency, Multi-Threading, Memory Management |

- Built a multithreaded Orderbook API that can add, remove, modify, or match market orders and FoK orders. Used heaps and time-based queues for fast order lookups and matches and memory pools to track lifetimes of orders. Achieved matching speeds of **5 nanoseconds per order**.